



CustomTees OMS

From Replit prototype to production on Google Cloud

Customer Order Management System · Leadership Demo · July 2026

Christy

What We Built

A production-ready internal tool for the whole order lifecycle

CustomTees OMS manages customers, orders, products with size and color variants, inventory, and analytics — all in one place, connected to a live PostgreSQL database with real-time stock deduction on every order.



Orders & Customers

Create orders, track status, manage customer records — stock auto-deducts on order creation.



Products & Inventory

Variant support (size / color), stock levels, and low-stock alerts with per-product thresholds.



Analytics

Sales overview, product performance, monthly revenue and purchase trends.



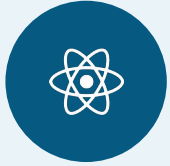
Users & Roles

Admin and employee roles, admin-only user management, JWT-secured sessions.

One codebase. Four modules. Two roles.

Under the Hood

A modern, single-language TypeScript stack



Frontend

React 19
TanStack Query
Tailwind CSS v4
shadcn/ui + Recharts



Backend

Express.js
REST API
JWT auth + bcrypt
Role middleware



Database

PostgreSQL
Drizzle ORM
Typed schema
Versioned migrations



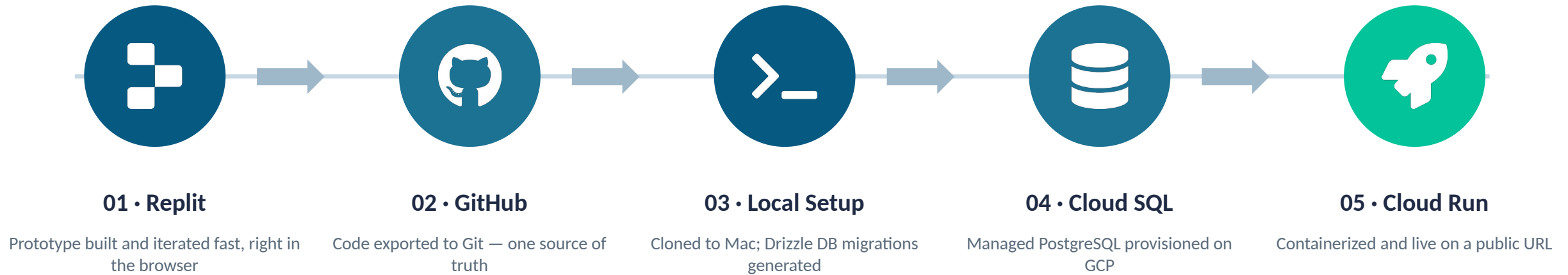
Build & Ship

Vite + TypeScript
Node 20
Docker
Cloud Build CI

97.5% TypeScript — one language across frontend, backend, and database schema

The Journey: Prototype → Production

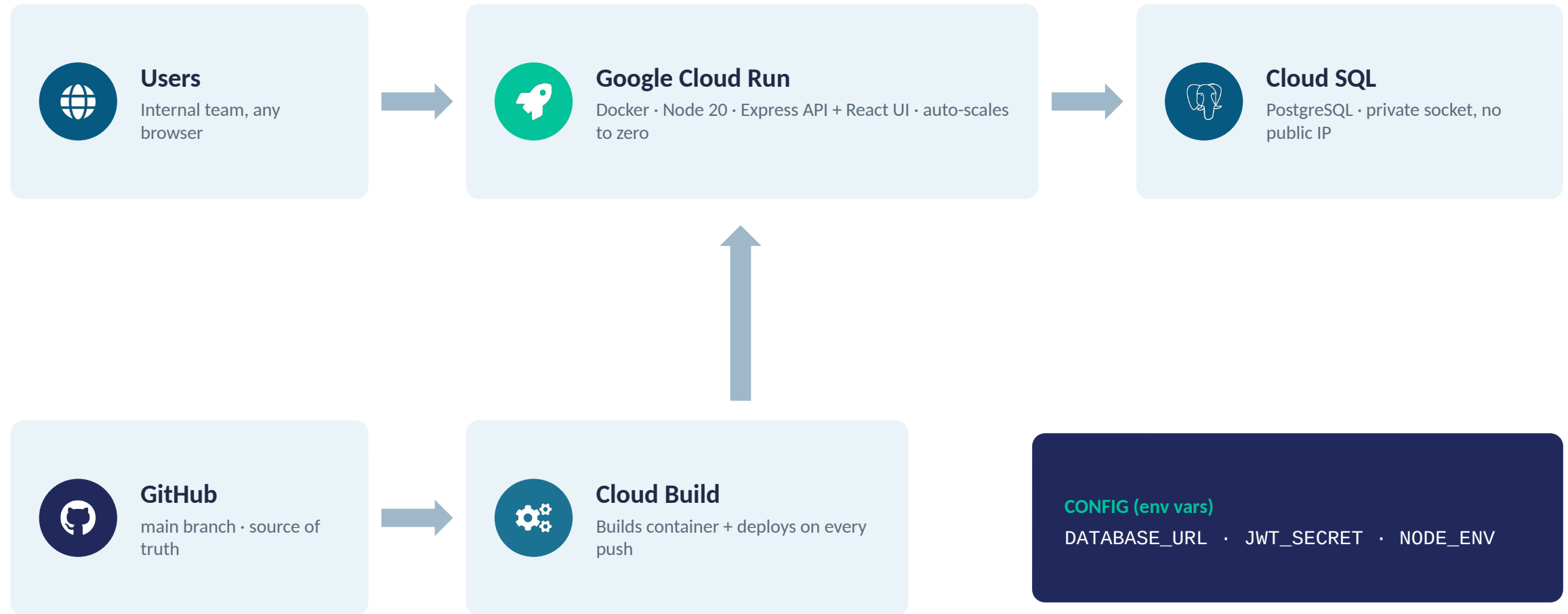
Five steps from idea to a live cloud service



Every step is repeatable — the app can be rebuilt from the repo alone

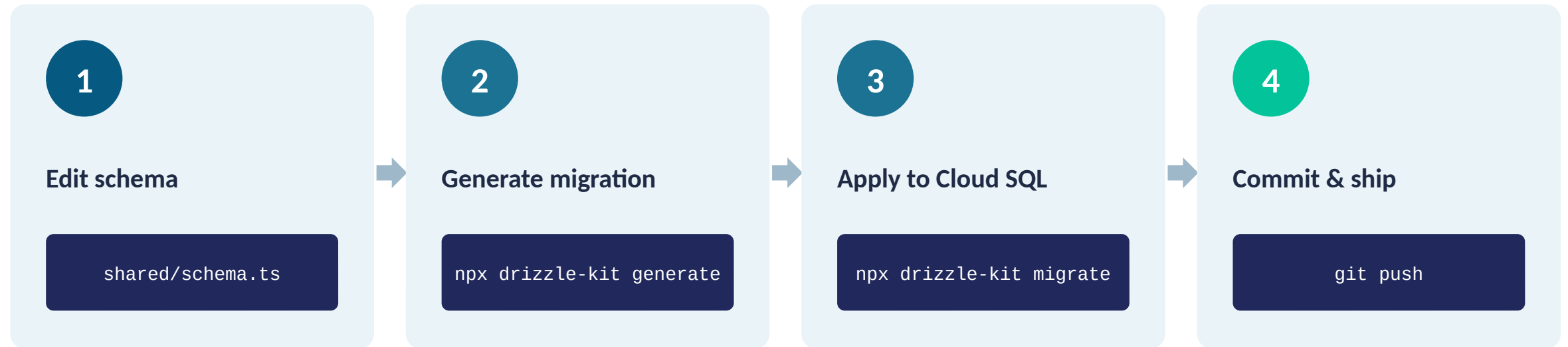
Production Architecture on Google Cloud

Serverless and fully managed — no servers to patch, pay only for use



Database Changes, Done Safely

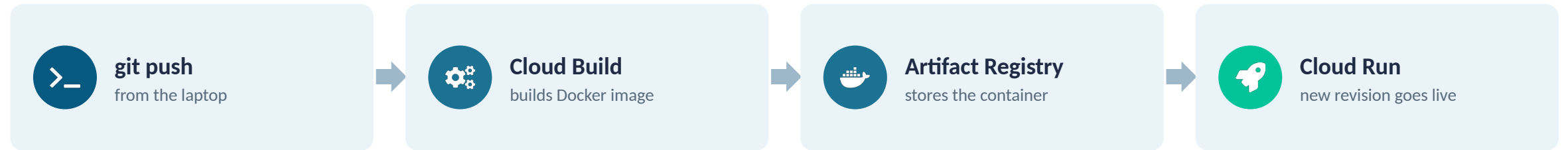
Drizzle ORM keeps schema and code in lockstep



Golden rule: migrations run before new app code deploys — database and repo never drift apart.

Push to Deploy

Continuous delivery with zero manual builds



One command to production: **git push**



Every push to main triggers a fresh build and deployment — no manual steps



Revisions roll out with zero downtime for users



Instant rollback — route traffic back to any previous revision

Security & Operations

Least privilege, encrypted paths, and no secrets in code



JWT authentication with bcrypt-hashed passwords



Role-based access — admin and employee permission levels



No public database IP in production — Cloud SQL socket only



Least-privilege IAM — the service gets only the Cloud SQL Client role



Secrets injected as environment variables — never committed to the repo



Security built in from day one — not bolted on later

What's Next

The platform is ready to grow with the business



Custom Domain

Branded URL with managed HTTPS certificates on Cloud Run.



Monitoring & Alerts

Cloud Logging dashboards, uptime checks, and error alerting.



Staging Pipeline

A separate staging service, with migrations automated in CI.



Feature Growth

Deeper analytics, exports, and integrations as team needs evolve.

Serverless foundation = new capabilities without new infrastructure



Let's See It Live

Live demo on Cloud Run — real data, real time

`github.com/ChristyDharmasingh/CustomTees`

Thank you — questions welcome